

HIGHER ORDER DERIVATIVES

	Function	1 st Derivative	2 nd Derivative	3 rd Derivative	n th Derivative
Lagrange	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(n)}(x)$
Leibniz	y	$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	$\frac{d^ny}{dx^n}$

Example 1: Determine the 2nd derivative of $f(x) = x^4 + 3x^2 - 5$.

$$f'(x) = 4x^3 + 6x$$

$$f''(x) = 12x^2 + 6$$

$$= 6(2x^2 + 1)$$

Example 2: Determine $\frac{d^2y}{dx^2}$ given that $y = \frac{3x-5}{2x}$.

$$\rightarrow y = \frac{3}{2} - \frac{5}{2x}$$

$$\frac{dy}{dx} = \frac{5}{2x^2}$$

$$\frac{d^2y}{dx^2} = -\frac{5}{x^3}$$

$$y = \frac{3}{2} - \frac{5x^{-1}}{2}$$

$$f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}}$$

Example 3: Determine $f'''(x)$ of the following functions.

a) $f(x) = \frac{1}{x}$

$$f'(x) = -x^{-2}$$

$$f''(x) = 2x^{-3}$$

$$f'''(x) = -6x^{-4}$$

$$f^{(4)}(x) = \frac{24}{x^5}$$

b) $f(x) = 4x^7 + 5x^6 - 8x^4 + 23x - \frac{\pi}{3}$

$$f'(x) = 28x^6 + 30x^5 - 32x^3 + 23$$

$$f''(x) = 168x^5 + 150x^4 - 96x^2$$

$$f'''(x) = 840x^4 + 600x^3 - 192x$$

$$f^{(4)}(x) = 3360x^3 + 1800x^2 - 192$$

c) $f(x) = \sqrt{x}$

$$f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$$

$$f''(x) = -\frac{1}{4}x^{-\frac{3}{2}}$$

$$f'''(x) = \frac{3}{8}x^{-\frac{5}{2}}$$

$$= \frac{3}{8\sqrt{x^5}}$$

Velocity and Acceleration

- **Position**, $s(t)$, is the location of an object at time, t .
- **Velocity**, $v(t)$, is the rate of change of position over time, so $v(t) = s'(t) = \frac{ds}{dt}$.
- **Acceleration**, $a(t)$, is the rate of change of velocity over time, so $a(t) = v'(t) = s''(t)$.

or $a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$.

$\{ s^{(4)}(t) | s^{(5)}(t) | s^{(6)}(t) \} \rightarrow \text{Snap, crackle, pop.}$

$a''(t) \rightarrow s'''(t)$ JERK

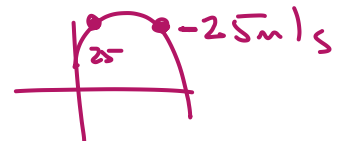
Example 1: The height of a soccer ball above the ground at time, t after it is kicked into the air, is given by the formula

$s(t)$ $\longrightarrow$ $h(t) = -5t^2 + 30t + 1$

where h is the height in metres, t is the time in seconds, and $t \geq 0$.

- a) Determine an equation that will calculate the instantaneous rate of change of the height of the ball with respect to time, t .

$v(t) = h'(t) = -10t + 30$



- b) Determine the velocity of the ball 0.5 seconds after it is kicked. At what other time will the ball have the same speed?

$\rightarrow v(0.5) = -10(0.5) + 30$

$v(0.5) = \underline{25 \text{ m/s}}$

$\left\{ \begin{array}{l} v(t) = -25 \text{ m/s.} \\ -10t + 30 = -25 \\ -10t = -55 \\ t = \underline{5.5 \text{ seconds}} \end{array} \right.$

- c) At what time will the ball reach its maximum height? What is the maximum height reached by the ball?

$m_t = 0$

max. height occurs at $v(t) = 0$ $\left\{ \begin{array}{l} h(3) = -5(3)^2 + 30(3) + 1 \\ h(3) = 46 \text{ m.} \end{array} \right.$

$v(t) = -10t + 30 = 0$

$t = \underline{3 \text{ seconds}}$

Vertex

$\rightarrow (3, 46)$

- d) What is the instantaneous velocity of the ball as it hits the ground?

$$h(t) = 0$$

$$t = \frac{-30 \pm \sqrt{900 + 20}}{-10}$$

inadmissible -
 $t > 0$

$$t \doteq 6.03 \text{ and } -0.033$$

$$\therefore v(6.03) \doteq -10(6.03) + 30$$

$$v(6.03) \doteq -30.3 \text{ m/s}$$

Example 2: A freight train leaves a train station and travels due north on a track. After t hours, the train is $s(t) = 18t^2 - 2t^3$, $0 \leq t \leq 9$ kilometres north of the train station.

- a) Determine an expression for the velocity of the train at any time $0 \leq t \leq 9$.

$$v(t) = s'(t)$$

$$v(t) = 36t - 6t^2, \quad 0 \leq t \leq 9$$

- b) Determine the velocity of the train after 3 hours and after 7 hours.

3 hrs

$$v(3) = 36(3) - 6(3)^2$$

$$v(3) = 108 - 54$$

$$v(3) = 54 \text{ km/hr}$$

7 hrs

$$v(7) = 36(7) - 6(7)^2$$

$$v(7) = 252 - 294$$

$$v(7) = -42 \text{ km/hr}$$

- c) Determine the acceleration of the train after 1 hour.

$$a(t) = v'(t)$$

$$a(t) = 36 - 12t$$

$$a(1) = 36 - 12(1)$$

$$a(1) = 24 \text{ km/hr}^2$$

km/hr/hr
 $= \text{km/hr}^2$

Additional Questions

1. Given $f(x) = ax^2 + bx + c$ and $f(1) = 8$, $f'(1) = 3$ and $f''(1) = -4$, determine the values of a , b and c .

$$* f'(x) = 2ax + b$$

$$* f''(x) = 2a \Rightarrow 2a = -4 \Rightarrow \boxed{a = -2}$$

$$* f'(x) = 2ax + b$$

$$f'(1) = 2(-2)(1) + b = 3$$

$$-4 + b = 3$$

$$\boxed{b = 7}$$

$$* f(1) = (-2)(1)^2 + 7(1) + c = 8$$

$$-2 + 7 + c = 8$$

$$\boxed{c = 3}$$

2. The position of a particle is given by $s(t) = 2t^3 - 39t^2 + 216t$, $t \geq 0$. Determine the acceleration of the particle at the instant when the velocity is zero.

$$v(t) = s'(t) = 6t^2 - 78t + 216$$

$$a(t) = v'(t) = 12t - 78$$

$$v(t) = 0$$

$$6t^2 - 78t + 216 = 0$$

$$t^2 - 13t + 36 = 0$$

$$(t-9)(t-4) = 0$$

$$t = 9 \text{ or } t = 4$$

$$a(9) = 12(9) - 78$$

$$* a(9) = 30 \text{ m/sec}^2$$

$$a(4) = 12(4) - 78$$

$$* a(4) = -30 \text{ m/sec}^2$$